COMPUTER VISION

LABSHEET 9

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# Experiment No. 9
# Object Detection and Recognition using Deep Learning Models

from ultralytics import YOLO
import cv2

# Load a pre-trained YOLO model
# The model will be downloaded automatically during the first run
model = YOLO("yolov8n.pt")

# Load an image
image_path = "image.jpg"
image = cv2.imread(image_path)

# Check whether the image was loaded
if image is None:
    print("Error: image.jpg was not found.")
    print("Place image.jpg in the same folder as this Python file.")
    exit()

# Perform object detection
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results = model(image)

# Display detected objects
for result in results:

    # Print detected class names and confidence scores
    if result.boxes is not None:
        for box in result.boxes:
            class_id = int(box.cls[0])
            confidence = float(box.conf[0])
            class_name = model.names[class_id]

            print(
                f"Object: {class_name}, "
                f"Confidence: {confidence:.2f}"
            )

    # Draw bounding boxes and labels
    annotated_image = result.plot()

# Display the result
cv2.imshow("YOLO Object Detection", annotated_image)

# Save the output image
cv2.imwrite("detected_objects.jpg", annotated_image)

print("Output saved as detected_objects.jpg")
print("Press any key to close the image window.")
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cv2.waitKey(0)

cv2.destroyAllWindows()


# Observation

# YOLO detects multiple objects in an image using bounding boxes and class labels.

# Confidence scores indicate the model's confidence in each prediction.

# The pre-trained model can detect common objects from the COCO dataset.

# Detection performance may be affected by image quality, object size, occlusion,

# lighting conditions, and background complexity.


# Question 1: Differentiate between image classification, object detection, and image segmentation.

# Image classification assigns one or more labels to an entire image.

# Object detection identifies objects using bounding boxes and class labels.

# Image segmentation assigns a class label to individual pixels or regions.


# Question 2: Explain the working principle of the YOLO algorithm.

# YOLO processes the complete image in a single forward pass through a neural network.

# It predicts bounding boxes, class probabilities, and confidence scores simultaneously.

# This makes YOLO suitable for real-time object detection.


# Question 3: Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and applications.

# YOLO is generally fast and suitable for real-time applications.

# SSD is also a single-stage detector with a balance between speed and accuracy.

# Faster R-CNN is a two-stage detector and usually provides high accuracy but is slower.


# Question 4: What is a bounding box? Why is it important?
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A bounding box is a rectangle surrounding a detected object.

It shows the object's location and helps separate it from the background.

Question 5: Explain confidence score and IoU.

A confidence score represents how confident the model is about a detection.

Intersection over Union measures the overlap between predicted and actual bounding boxes.

Higher IoU indicates better localization.

Question 6: What are the advantages of pre-trained deep learning models?

Pre-trained models reduce training time, require less data, and provide strong features.

They are generally more effective than manually designed image-processing rules.

Question 7: Why are MS COCO and Pascal VOC widely used?

They contain many images, object categories, annotations, and bounding boxes.

They provide standard benchmarks for evaluating object detection models.

Question 8: Mention five real-world applications.

Applications include autonomous driving, surveillance, healthcare, retail analytics,

and industrial inspection.

Question 9: What challenges occur in complex environments?

Challenges include occlusion, poor lighting, small objects, overlapping objects,

motion blur, cluttered backgrounds, and changes in object appearance.

Question 10: How can object detection be improved?

Performance can be improved using better datasets, data augmentation, transfer learning,

larger models, hyperparameter tuning, improved feature extraction, and model pruning.

OUTPUT

